

■ Scientific Justification

Circumstellar disks of gas and dust are a natural by-product of the star formation process and provide the raw material for planet formation. Hence, their evolution and dispersal determine when and what types of planets can form. *Here, we propose to use MIRI MRS to map the winds from two disks with large dust cavities, constrain the ionization fraction in the flow, and wind mass loss rates. Our observations and analysis will establish how much time is left in these systems to complete planet formation and migration.*

Disk evolution and dispersal timescales. Infrared and millimeter surveys of nearby star-forming regions have established that the typical protoplanetary disk lifetime is a few Myr but the disk dispersal timescale, i.e. the transition from disk-bearing to disk-less, is an order of magnitude shorter (e.g., Ercolano & Pascucci 2017 for a recent review). While the relative roles of viscous accretion vs. magnetohydrodynamic (MHD) disk winds in driving evolution is debated (e.g., Turner et al. 2014), *photoevaporation driven by high-energy stellar photons from the central star* has been long recognized as critical to the dispersal process (Clarke et al. 2001). According to the classic evolutionary picture (e.g., Alexander et al. 2014 and Figure 1), viscous accretion slowly depletes disk material and when the disk accretion rate falls below the photoevaporative wind loss rate, photoevaporation takes over and quickly clears the disk. Thus, photoevaporation effectively ends giant planet formation (e.g. Shu et al. 1993) and limits planet migration which, in turn, impacts the final architecture of planetary systems (e.g., Alexander & Pascucci 2012, Jennings et al. 2018). *Our proposed observations have the potential to provide the first maps of these photoevaporative winds.*

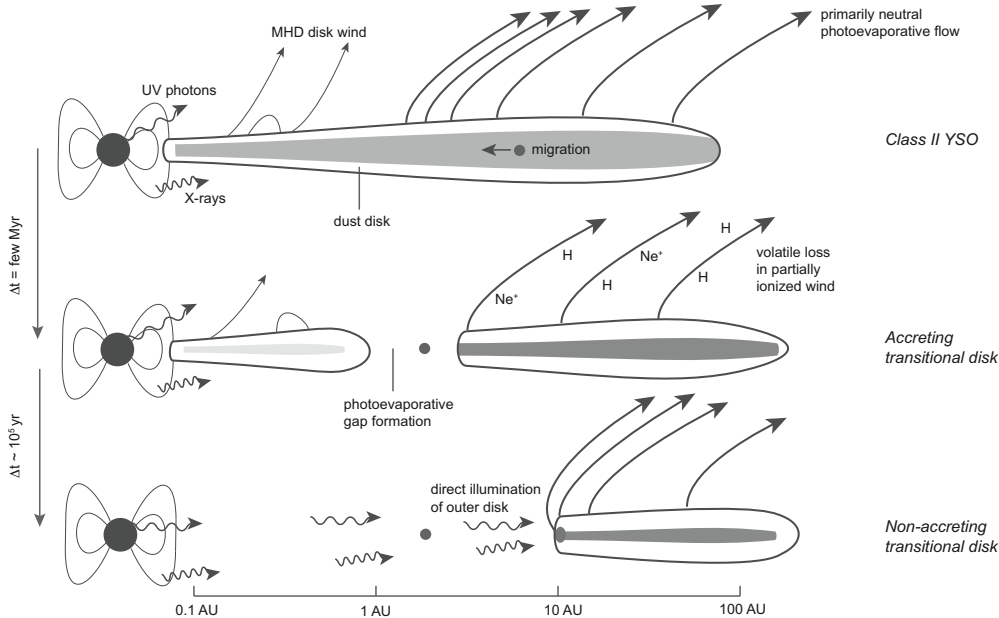


Figure 1: Schematic representation of the classic evolutionary picture of disk evolution and dispersal (Alexander et al. 2014).

Cosmic neon lights the way is the title of the *Spitzer* Science Center press release announcing the discovery of ionized neon towards isolated low-mass stars with disks (Espaillat et al. 2007, Lahuis et al. 2007, Pascucci et al. 2007). These [Ne II] 12.81 μm detections led to